Stat 286: Causal Inference with Applications, Fall 2025

Section 2: Permutation Test

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1 Overview

Setup:

- Units: $i = 1, \dots, n$
- Treatment: $T_i \in \{0, 1\}$
- Outcome: $Y_i = Y_i(T_i)$
- Finite-Population framework: potential outcomes $\mathcal{O}_n = \{Y_i(0), Y_i(1)\}_{i=1}^n$ are fixed and the only randomness is the assignment mechanism T.
- Completely Randomized Design (CRD): $\Pr(T_i = 1 \mid \mathcal{O}_n) = \Pr(T_1 = 1) = \frac{n_1}{n}$

Goal: Test for

1. Sharp Null: $H_0(\tau_0) : Y(1) - Y_i(0) = \tau_0 \quad \text{for all } i$
2. Maximal Effect: $H_0^{\max}(\tau_0) : Y_i(1) - Y_i(0) \leq \tau_0 \quad \text{for all } i$
3. Quantile Effect: $H_0^{\alpha}(\tau_0) : Q_{Y_i(1)-Y_i(0)}(\alpha) \leq \tau_0 \quad \text{for all } i$; $Q_X(\alpha) = \inf\{x \in \mathbb{R} : \Pr(X \leq x) \geq \alpha\}$ (for α percent of the population the treatment effect is smaller than τ_0).

2 Fisher's Sharp Null

We test the sharp null of constant treatment effect,

$$H_0(\tau_0) : \quad Y_i(1) - Y_i(0) = \tau_0 \quad \forall i.$$

There are several implications when we consider the special case $\tau_0 = 0$. In this case the null assumes that the population only consists of Always-Takers and Never-Takers. There is an interesting paper [Christy and Kowalski, 2025] that provides an alternative in terms of how the strata distribution looks like when $H_0(0)$ is rejected. It furthers implies that there are no spillover effects within the population.

2.1 Permutation Tests

Define a test statistic based on observables $S_\tau = f(\boldsymbol{T}, \boldsymbol{Y}, \tau)$ that you believe has power against the null. For example, use

$$S_{\tau_0} = \frac{1}{n_1} \sum_{i=1}^{n} T_i \, Y_i(T_i) \; - \; \frac{1}{n_0} \sum_{i=1}^{n} (1 - T_i) \, (Y_i(T_i) + \tau_0)$$

There are many possibilities, but calibrating the test statistic to $\mathbb{E}[S_{\tau_0} \mid H_0] = 0$ makes inference easier. This is a special case of an important *difference-in-means* class of test statistics, defined by

$$S_{\tau_0, \phi_1, \phi_0} = \sum_{i=1}^{n} T_i \phi_1(Y_i(T_i)) - \sum_{i=1}^{n} (1 - T_i) \, \phi_0(Y_i(T_i))$$

with $\phi_1(y) = \frac{y}{n_1}$ and $\phi_0(y) = \frac{y + \tau_0}{n_0}$.

Given a test statistic, we want to test it against the null, i.e., obtain valid p-values, confidence intervals and a point estimate for τ.

The basis for all of that is to perform a so called permutation test:

- Draw $b = 1, \dots, B$ random assignments $\boldsymbol{T}^{(b)} \in \{0,1\}^n$.
- Compute potential outcomes under each assignment:
  - If $T_i^{(b)} = T_i$: $Y_i(T_i^{(b)}) = Y_i(T_i)$
  - If $(T_i^{(b)}, T_i) = (0,1)$: $Y_i(T_i^{(b)}) = Y_i(0) = Y_i(1) - \tau_0 = Y_i(T_i) - \tau_0$
  - If $(T_i^{(b)}, T_i) = (1,0)$: $Y_i(T_i^{(b)}) = Y_i(1) = Y_i(0) + \tau_0 = Y_i(T_i) + \tau_0$
- Recalculate the test statistic:

$$S_{\tau_0}^{(b)} = \frac{1}{n_1} \sum_{i=1}^{n} T_i^{(b)}\, Y_i(T_i^{(b)}) - \frac{1}{n_0} \sum_{i=1}^{n} (1 - T_i^{(b)})\, (Y_i(T_i^{(b)}) + \tau_0).$$

P-values. We then calculate the p-values depending on the alternative.

$$H_1 : \ \tau \neq \tau_0 \quad \Rightarrow \quad p_{\text{perm}}(\tau_0) = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\{|S^{(b)} - \mu| \geq |S_{\tau_0}^{\text{obs}} - \mu|\}}{B+1}.$$

$$H_1 : \ \tau > \tau_0 \quad \Rightarrow \quad p_{\text{perm},+}(\tau_0) = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\{S^{(b)} \geq S_{\tau_0}^{\text{obs}}\}}{B+1}.$$

$$H_1 : \ \tau < \tau_0 \quad \Rightarrow \quad p_{\text{perm},-}(\tau_0) = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\{S^{(b)} \leq S_{\tau_0}^{\text{obs}}\}}{B+1}.$$

where $\mu = \mathbb{E}[S_{\tau_0} \mid H_0(\tau_0)]$; here $\mu = 0$. Alternatively, the two sided p-values can also be obtain from the one-sided p-values

$$p_{\text{perm}}(\tau_0) = 2 \cdot \min(p_{\text{perm},+}(\tau_0), p_{\text{perm},-}(\tau_0)).$$

Normal Approximation Some test statistics, like the one above, can be approximated by a normal distribution. Indeed, we will show in the next module that under the null $H_0(\tau_0)$ we have

$$Z = \frac{\sqrt{n}(S_{\tau_0}^{\text{obs}} - \mu)}{\sqrt{\sigma_n^2}} \xrightarrow{d} \mathcal{N}(0,1),$$

where $\mu = \mathbb{E}[S_{\tau_0}^{\text{obs}} \mid H_0(\tau_0)]$ (here $\mu = 0$) and $\sigma_n^2 = S^2 \left(\frac{1}{n_1} + \frac{1}{n_0}\right)$, $S^2 \equiv S(t)^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(Y_i(t) - \overline{Y}(t)\right)^2$ for $t \in \{0,1\}$ under H_0 (which can be estimated by $\hat{\sigma}_t = \frac{1}{n_t - 1} \sum_{i:T_i = t} (Y_i - \overline{Y}_t)^2$). P-values can then be obtained by

$$H_1 : \ \tau \neq \tau_0 \quad \Rightarrow \quad p_{\text{norm}}(\tau_0) = 2\{1 - \Phi(|Z|)\}.$$

$$H_1 : \ \tau > \tau_0 \quad \Rightarrow \quad p_{\text{norm},+}(\tau_0) = 1 - \Phi(Z),$$

$$H_1 : \ \tau < \tau_0 \quad \Rightarrow \quad p_{\text{norm},-}(\tau_0) = \Phi(Z),$$

where Φ is the CDF of the standard normal distribution.

Confidence set. We can obtain a $(1-\alpha)$ confidence set by inverting the test,

$$\mathcal{C}_{1-\alpha} = \{\tau : \ p(\tau) > \alpha\},$$

where $p(\tau)$ is the p–value for $H_0(\tau)$. If the test statistic is monotone in τ, then $\mathcal{C}_{1-\alpha}$ is an interval. In practice, evaluate $p(\tau)$ over a grid and keep all τ with $p(\tau) > \alpha$.

Point estimate.

$$\hat{\tau} = \arg\max_{\tau} \ p(\tau).$$

Permutation Inference vs. Normal Approximation.

Permutation Inference

- Pro: Sample size n can be small (no large-sample assumption), for $B \to \infty$ recovers *exact* p-values (under the sharp null)
- Cons: Computational cost grows with B, especially when inverting the test

Normal Approximation

- Pro: Very fast, closed form p-values formula
- Cons: Need large sample size n for the asymptotics to kick in

Pros and Cons of the Difference-in-Means Test Statistics

- **Pros**: Y_i can be binary, categorical, or continuous; also work for the maximal hypothesis $H_0^{\max}(\tau_0)$
- **Cons**: Can be less powerful with heavy tails; sensitive to outliers. Not distribution-free (test statistic scales with the scale of the outcomes, depends on the actual outcome values), so it cannot test quantile hypothesis $H_0^{\alpha}(\tau_0)$

This motivates looking at rank score tests, for example the Wilcoxon's rank-sum test.

2.2 Wilcoxon's rank-sum test

In this section we require non-binary outcomes $Y_i \notin \{0,1\}$ or count outcomes with many ties. Often the rank sum tests are used for continuous outcomes. We still assume the sharp null, but now the test statistic is given by

$$S_\tau \;=\; \sum_{i=1}^{n} T_i \, R_i\big(\boldsymbol{Y} - \tau \boldsymbol{T}\big),$$

with $R_i(\cdot)$ the rank among all n adjusted outcomes (average ranks for ties). This test statistic has important characteristics.

Invariance under H_0: Let $\boldsymbol{T}, \boldsymbol{A} \in \{0,1\}^n$ be two treatment assignments. Under $H_0(\tau_0)$ we have

$$\boldsymbol{Y}(\boldsymbol{T}) - \tau_0 \boldsymbol{T} \;=\; \boldsymbol{Y}(0) \;=\; \boldsymbol{Y}(\boldsymbol{A}) - \tau_0 \boldsymbol{A},$$

since

- $T_i = 1:\; Y_i(T_i) - \tau_0 T_i = Y_i(1) - \tau_0 = Y_i(0)$,
- $T_i = 0:\; Y_i(T_i) - \tau_0 T_i = Y_i(0)$.

Thus, under $H_0(\tau_0)$

$$S_{\tau_0} = \sum_{i=1}^{n} T_i \, R_i\big(\boldsymbol{Y} - \tau_0 \boldsymbol{T}\big) = \sum_{i=1}^{n} T_i \, R_i(\boldsymbol{Y}(0)) = \sum_{i=1}^{n} A_i \, R_i(\boldsymbol{Y}(0)),$$

does not depend on τ_0. This facilitates inference, as we will later see.

Monotonicity S_τ is nonincreasing in τ; if $\tau_1 < \tau_2$ then $S_{\tau_1} > S_{\tau_2}$.

Inference We can also perform a permutation test for the Wilcoxon's rank-sum test

- Draw $b = 1, \ldots, B$ random assignments $\boldsymbol{T}^{(b)} \in \{0,1\}^n$.
- Recalculate the test statistic:

$$S_{\tau_0}^{(b)} = \sum_{i=1}^{n} T_i^b \, R_i\big(\boldsymbol{Y} - \tau \boldsymbol{T}\big)$$

where we can use the same ranks for every draw and don't have to adjust whats inside the rank by the invariance property.

Normal-Approximation If n is large, a permutation test will take a lot of time. Alternatively, we can perform a normal approximation. The idea is that under $H_0(\tau_0)$ by the invariance property, the distribution of S_{τ_0} does not depend on τ_0! Even without H_0 by the robustness of ranks the mean and variance don't depend on τ

$$\mathbb{E}[S_\tau \mid \mathcal{O}_n] = \frac{n_1(n+1)}{2}, \qquad \mathrm{Var}(S_\tau \mid \mathcal{O}_n) = \frac{n_0 n_1(n+1)}{12}.$$

To obtain p-values compute

$$Z = \frac{S_{\tau_0}^{\mathrm{obs}} - E[S_\tau \mid \mathcal{O}_n]}{\sqrt{\mathrm{Var}(S_\tau \mid \mathcal{O}_n)}}$$

$$\begin{aligned} H_1 : \ \tau \neq \tau_0 \quad &\Rightarrow \quad p_{\mathrm{norm}}(\tau_0) = 2\{1 - \Phi(|Z|)\}. \\ H_1 : \ \tau > \tau_0 \quad &\Rightarrow \quad p_{\mathrm{norm},+}(\tau_0) = 1 - \Phi(Z), \\ H_1 : \ \tau < \tau_0 \quad &\Rightarrow \quad p_{\mathrm{norm},-}(\tau_0) = \Phi(Z). \end{aligned}$$

We can then invert these p-values to obtain a confidence interval (since S_τ is monotone) and a point estimate.

3 Maximal Individual Effect

The sharp null is very strict and does not allow for treatment effect heterogeneity. However, we can extend the framework to test for the largest unit-level effect [Caughey et al., 2023],

$$H_0^{\max}(\tau_0) : \max_{1 \le i \le n} \{Y_i(1) - Y_i(0)\} \ \le \ \tau_0.$$

If the outcomes are binary then $\tau_0 = 1$ is trivially true. Thus we focus on the non binary outcome case.

Insight. Denoting the Wilcoxon rank-sum test statistic for τ_0 as S_{τ_0} (it does not have to be the Wilcoxon test statistic, difference in means and other rank sum tests also work) we have

$$\Pr(S_{\tau_0} \ge s \mid H_0^{\max}(\tau_0)) \le \Pr(S_{\tau_0} \ge s \mid H_0(\tau_0)).$$

Thus if we reject the same test statistic under the one-sided sharp null

$$p(\tau_0) := \Pr(S_{\tau_0} \ge S_{\tau_0}^{\mathrm{obs}} \mid H_0(\tau_0)) \le \alpha,$$

with the methods we discussed above, we also safely reject the maximal hypothesis. This yields a valid, but conservative test.

Confidence Interval We can form a corresponding confidence interval by again setting

$$\mathcal{C}_{1-\alpha}^{\max} \ = \ \{\tau : \ p(\tau) > \alpha\},$$

which for our test statistics will always be an interval of the form $[\underline{\tau}^{\max}, \infty)$ with $\underline{\tau}^{\max} = \inf\{\tau : p(\tau) > \alpha\}$. We can do the same in reverse to obtain a $1-\alpha$ interval for the minimum effect $(-\infty, \overline{\tau}^{\min}]$.

To obtain a two sided interval we first obtain $1 - \frac{\alpha}{2}$ confidence intervals $[\underline{\tau}^{\max}_{1-\frac{\alpha}{2}}, \infty)$ and $(-\infty, \overline{\tau}^{\min}_{1-\frac{\alpha}{2}}]$. A valid $1-\alpha$ confidence interval is then given by

$$[\max(\underline{\tau}^{\max}_{1-\frac{\alpha}{2}} - \overline{\tau}^{\min}_{1-\frac{\alpha}{2}}, 0), \infty]$$

Proof. We limit ourselves to test statistics of the form $S_{\tau_0}(\boldsymbol{T},\boldsymbol{Y}) = S(\boldsymbol{T},\boldsymbol{Y}-\tau_0\boldsymbol{T})$. We say that a function $S(\boldsymbol{T},\boldsymbol{Y})$ is *effect increasing*, if

$$S(\boldsymbol{T},\boldsymbol{Y}+\boldsymbol{T}\boldsymbol{\eta}+(1-\boldsymbol{T})\boldsymbol{\zeta}) \geq S(\boldsymbol{T},\boldsymbol{Y}),$$

for $\boldsymbol{\eta} \geq 0, \boldsymbol{\zeta} \leq 0$. Note that the Wilcoxon rank-sum test statistic is both of the desired functional form ($S_{\tau_0}(\boldsymbol{T},\boldsymbol{Y}) = S(\boldsymbol{T},\boldsymbol{Y}-\tau_0\boldsymbol{T}) = \sum_{i=1}^n T_i R_i(\boldsymbol{Y}-\tau_0\boldsymbol{T})$) and is effect increasing as for each i

$$T_i R_i(\boldsymbol{Y}+\boldsymbol{T}\boldsymbol{\eta}+(1-\boldsymbol{T})\boldsymbol{\zeta}) \geq T_i R_i(\boldsymbol{Y})$$

as we only count the rank of the treated units which are elevated by $\boldsymbol{\eta}$ while the control units (which we don't count) are decreased by $\boldsymbol{\zeta}$. Denote with $\tau_i = Y_i(1) - Y_i(0)$ the individual treatment effect. Note that

$$Y_i = Y_i(T_i) = Y_i(1)T_i + Y_i(0)(1-T_i) = Y_i(0) + \tau_i T_i$$

Then for two different treatments we have $T_i, A_i \in \{0,1\}$ we have

$$\begin{aligned}(Y_i(T_i)-\tau_0 T_i) - (Y(A_i)-\tau_0 A_i) &= Y_i(0) + (\tau_i-\tau_0)T_i - Y_i(0) - (\tau_i-\tau_0)A_i \\ &= (\tau_i-\tau_0)T_i + (\tau_0-\tau_i)A_i \\ &= (\tau_i-\tau_0)T_i + (\tau_0-\tau_i)A_i + A_iT_i(\tau_i-\tau_0) + A_iT_i(\tau_0-\tau_i) \\ &= (\tau_0-\tau_i)A_i(1-T_i) + (\tau_i-\tau_0)(1-A_i)T_i.\end{aligned}$$

Under $H_0^{\max}(\tau_0)$ we have $\eta_i = (1-T_i)(\tau_0-\tau_i) \geq 0$ and $\zeta_i = T_i(\tau_i-\tau_0) \leq 0$ (under the sharp null both would be zero). Assuming an effect-increasing test-statistic $S(\boldsymbol{T},\boldsymbol{Y})$, this implies that

$$\begin{aligned}S(\boldsymbol{A},\boldsymbol{Y}(\boldsymbol{T})-\tau_0\boldsymbol{T}) &= S(\boldsymbol{A},\boldsymbol{Y}(\boldsymbol{A})-\tau_0\boldsymbol{A}+\boldsymbol{A}\boldsymbol{\eta}+(1-\boldsymbol{A})\boldsymbol{\zeta}) \\ &\geq S(\boldsymbol{A},\boldsymbol{Y}(\boldsymbol{A})-\tau_0\boldsymbol{A}).\end{aligned}$$

Finally note that

$$\begin{aligned}\Pr(S(\boldsymbol{A},\boldsymbol{Y}(\boldsymbol{A})-\tau_0\boldsymbol{A}) \geq s \mid H_0^{\max}(\tau_0)) &= \sum_{\boldsymbol{a}} \Pr(\boldsymbol{A}=a)\mathbb{1}_{\{S(\boldsymbol{a},\boldsymbol{Y}(\boldsymbol{a})-\tau_0\boldsymbol{a})\geq s\}} \\ &\leq \sum_{\boldsymbol{a}} \Pr(\boldsymbol{A}=a)\mathbb{1}_{\{S(\boldsymbol{a},\boldsymbol{Y}(\boldsymbol{T})-\tau_0\boldsymbol{T})\geq s\}} \\ &= \Pr(S(\boldsymbol{A},\boldsymbol{Y}(\boldsymbol{T})-\tau_0\boldsymbol{T}) \geq s \mid H_0(\tau_0)) \\ &= \Pr(S(\boldsymbol{A},\boldsymbol{Y}(\boldsymbol{A})-\tau_0\boldsymbol{A}) \geq s \mid H_0(\tau_0))\end{aligned}$$

The first equality follows from the finite population assumption where only the treatment is randomized. The second equality follows from $S(\boldsymbol{a},\boldsymbol{Y}(\boldsymbol{T})-\tau_0\boldsymbol{T}) \geq S(\boldsymbol{a},\boldsymbol{Y}(\boldsymbol{a})-\tau_0\boldsymbol{a})$ under $H_0^{\max}(\tau_0)$. The third equality from the finite population assumption as well and from the fact that under $H_0(\tau_0) : \boldsymbol{Y}(\boldsymbol{T})-\tau_0\boldsymbol{T} = \boldsymbol{Y}(0)$ is independent of $\boldsymbol{T}$ and thus we don't need to sum over its randomization. The fourth equality follows since under $H_0(\tau_0) : \boldsymbol{Y}(\boldsymbol{T})-\tau_0\boldsymbol{T} = \boldsymbol{Y}(0) = \boldsymbol{Y}(\boldsymbol{A})-\tau_0\boldsymbol{A}$. This proves the claim.

4 Quantile Treatment Effects

We can even go a step further and test about the quantiles of the treatment effect, i.e.,

$$H_0^\alpha(\tau_0) : Q_{Y_i(1)-Y_i(0)}(\alpha) \leq \tau_0 \quad \text{for all } i; Q_X(\alpha) = \inf\{x \in \mathbb{R} : \Pr(X \leq x) \geq \alpha\}.$$

In our finite population context with n units this means testing the null hypothesis

$$H_0^k(\tau_0) : \tau_{(k)} \leq \tau_0,$$

where we have sorted the individual treatment effects $\tau_{(1)} \leq \tau_{(2)} \leq \ldots \leq \tau_{(n)}$. Let $\boldsymbol{\tau} = (\tau_1,\ldots,\tau_n)$.

Observe that

$$H_0^k(\tau_0) : \tau_{(k)} \leq \tau_0 \Leftrightarrow \boldsymbol{\tau} \in \mathcal{H}^k(\tau_0) := \left\{\boldsymbol{\delta} \in \mathbb{R}^n : \delta_{(k)} \leq \tau_0\right\} = \left\{\boldsymbol{\delta} \in \mathbb{R}^n : \sum_{i=1}^n \mathbb{1}_{\{\delta_i \geq \tau_0\}} \leq n-k\right\}$$

Intuitively, we are testing a maximal treatment hypothesis as before on the k-smallest individual treatment effects, while leaving the remaining $n-k$ as free parameters.

We continue to use the Wilcoxon rank-sum test statistic $S_{\boldsymbol{\tau}}$, but we allow the τ_i to vary in each entry. So how do we obtain a p-value? Recall that for a valid test at level α we can only reject the null if the probability that the null is true is α (so small). Given the equivalence of the null above, that means that we want a test statistic that only rejects if the event $\boldsymbol{\tau} \in \mathcal{H}^k(\tau_0)$ has low probability, or,

$$\forall \boldsymbol{\tau} \in \mathcal{H}^k(\tau_0) : \Pr(S_{\boldsymbol{\tau}} \geq S_{\boldsymbol{\tau}}^{\text{obs}}) \leq \alpha.$$

This is equivalent to

$$\sup_{\boldsymbol{\tau} \in \mathcal{H}^k(\tau_0)} p(\boldsymbol{\tau}) := \sup_{\boldsymbol{\tau} \in \mathcal{H}^k(\tau_0)} \Pr(S_{\boldsymbol{\tau}} \geq S_{\boldsymbol{\tau}}^{\text{obs}})$$

This probability can be upper bounded by choosing the smallest possible test statistic $\inf_{\boldsymbol{\tau} \in \mathcal{H}^k(\tau_0)} S_{\boldsymbol{\tau}}^{\text{obs}}$. So how do we choose such a minimizing $\boldsymbol{\tau}$?

Assume we are interested in the $1-\alpha$ quantile. First, set $k = \lceil n(1-\alpha) \rceil$, which means that our null does not restrict the remaining $n-k = \lfloor \alpha n \rfloor$ units. To minimize our statistic, we choose among the $n-k$ largest outcome values the treated units and set their maximal effect to ∞. One can prove that this is optimal. In other words,

$$\tau_i^* = \begin{cases} \infty, & T_i = 1, R_i(\boldsymbol{Y}) \geq \min(n-k, n_1) \\ \tau_0 \end{cases}$$

The intuition is as follows. We want to minimize our test statistic $S_{\boldsymbol{\tau}}^{\text{obs}} = \sum_{i=1}^{n} T_i R_i(\boldsymbol{Y} - \boldsymbol{\tau}\boldsymbol{T})$. We make the observations

1. In order to keep with the $\mathcal{H}^k(\tau_0)$ we need to set at least k values of $\boldsymbol{\tau}$ to τ_0
2. In our test statistic $S_{\boldsymbol{\tau}}^{\text{obs}}$ we only sum over the ranks of the treated units, so we want to make these ranks small, while still keeping with the null constrain
3. The smallest we can make them is by taking the largest $n-k$ $\boldsymbol{Y}$ entries and subtract the largest possible value from them.
4. The n_1 is a technical condition to ensure we set enough treated units to ∞

The test can then be inverted to obtain a confidence interval by $\mathcal{C}_{1-\alpha} = \{\tau_0 : \sup_{\boldsymbol{\tau} \in \mathcal{H}^k(\tau_0)} p(\boldsymbol{\tau}) \geq \alpha\}$.

References

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